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School of Computing and Mathematical Sciences

CO7201 Individual Project

Interim Report

**CareMate+: A Secure Android Healthcare Companion for
Medication Management, Remote Monitoring, Pharmacy
Fulfilment and Care Coordination**

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Table of Contents

1. Project progress update	4
1.1 Overall Implementation Status	4
1.2 Response to Preliminary Feedback	4
2. CareMate+ Implementation Progress and Requirement Traceability Against the Preliminary Report...	5
2.1 Essential Security and Patient Requirements	5
2.2 Essential Doctor and Administration Requirements	6
2.3 Essential Caregiver and Pharmacy Requirement Progress	6
2.4 Recommended and Optional Requirement Progress	7
2.5 Overall Requirement Status Summary	7
3. Technical Achievement Snapshot	8
3.1 Key MSc-Level Technical Contributions	8
3.2 Architecture, Integration and Development Evidence	8
4. Revisions to the Preliminary Plan and Evaluation Progress	9
4.1 Scope and Priority Changes	9
4.2 Technical Challenges and Engineering Decisions	9
4.3 Evaluation Completed So Far	10
4.4 Ethics and Remaining Evaluation Work	10
5. Revised Project Plan	10
5.1 Comparison with the Original Schedule	10
5.2 Updated Twelve-Week Schedule	11
5.3 Remaining Milestones and Risk Mitigation	12

List of Figures

Figure 1: Current implemented end-to-end CareMate+ workflow.....	5
Figure 2 : CareMate+ Core Role, Security and Data Architecture.....	9
Figure 3: Revised twelve-week CareMate+ project plan (Gantt chart) showing the original schedule, actual progress and revised delivery plan.	11

List of Tables

Table 1: Preliminary Feedback and Development Response	4
Table 2: Essential Security and Patient Requirement Progress	6
Table 3: Essential Doctor and Administration Requirement Progress.....	6
Table 4: Essential Caregiver and Pharmacy Requirement Progress	6
Table 5: Recommended Requirement Progress	7
Table 6: Optional Requirement Progress	7
Table 7: Overall Requirement Status Summary	7
Table 8: Key MSc-Level Technical Contributions	8
Table 9 : Original and Actual Schedule Comparison	10
Table 10: Revised Weekly Implementation and Delivery Plan	12
Table 11: Remaining Project Risks and Controls	12

1. Project progress update

1.1 Overall Implementation Status

CareMate+ has progressed from a proposed multi-role concept into an integrated Android healthcare-support system. The React Native application, Node.js/Express TypeScript API and PostgreSQL database managed through Prisma now support the principal patient, doctor and administrator workflows.

The patient module includes secure authentication, dashboard summaries, medicine and reminder management, taken, missed and snoozed dose tracking, OCR-assisted medicine and prescription scanning, editable confirmation, simulated vital readings and Android Health Connect synchronisation. Stable, warning and critical readings are classified automatically. Critical readings activate the Safety Response workflow, which provides a 30-second countdown, false-alarm cancellation and escalation to an assigned doctor through a linked Safety Alert and emergency consultation.

The doctor module supports professional approval, multiple patient assignments, assigned-patient access control, clinical record review, medicine-review decisions, doctor notes, alert management, manual and emergency consultations, Jitsi/JaaS video calls and manual or OCR-assisted prescription creation. Confirmed prescriptions generate the related medicine and reminder records, while consultation completion synchronises both devices and resolves the linked Safety Alert. The administrator module supports professional-document review and account approval or rejection.

Patient, doctor and administrator modules were prioritised and substantially implemented first. Caregiver and pharmacy development will begin after these core workflows have been fully integrated and verified. Firebase push notifications are currently ongoing as an extension to the Safety Response workflow.

The ethics documentation required for the planned participant evaluation has also been prepared. Minor amendments requested by the supervisor are currently being incorporated, and participant evaluation will begin only after ethics approval has been confirmed.

The remaining work focuses on push-notification integration, caregiver and pharmacy implementation, systematic evaluation, defect correction and final evidence preparation. CareMate+ remains a prototype and does not replace professional medical advice or emergency services.

1.2 Response to Preliminary Feedback

Preliminary feedback highlighted two risks: the advanced contribution needed clearer definition, and developing five equally detailed roles could make the project infeasible. The project priorities were therefore revised without changing the overall CareMate+ vision.

Feedback	Response and current outcome
Advanced contribution required clarification.	Technical depth was focused on editable OCR, relationship-based access control, Safety Response state management, prescription-to-reminder integration and cross-device consultation synchronisation.
Data visualisation alone is not an advanced feature.	Charts are treated as presentation layers; the main contribution is secure data acquisition, classification, persistence and role-controlled retrieval.
Five equally detailed roles may be unrealistic.	Patient, doctor and administrator modules were prioritised as the core pathway. Caregiver and pharmacy development will begin after these modules are fully integrated and verified.
Scope required tighter control.	Firebase push notifications remain ongoing, while supporting-role implementation, testing and defect correction are scheduled for the remaining period.
Human-participant evaluation requires ethics approval.	The participant information sheet, consent form, study protocol and questionnaire have been prepared. After supervisor-requested amendments, the Infonetica form will be completed and all documents uploaded for review. Participant evaluation will begin only after approval.

Table 1: Preliminary Feedback and Development Response

This approach protects the quality of the central patient-doctor workflow while retaining the wider five-role objective. Human-participant evaluation will follow the required ethics process.

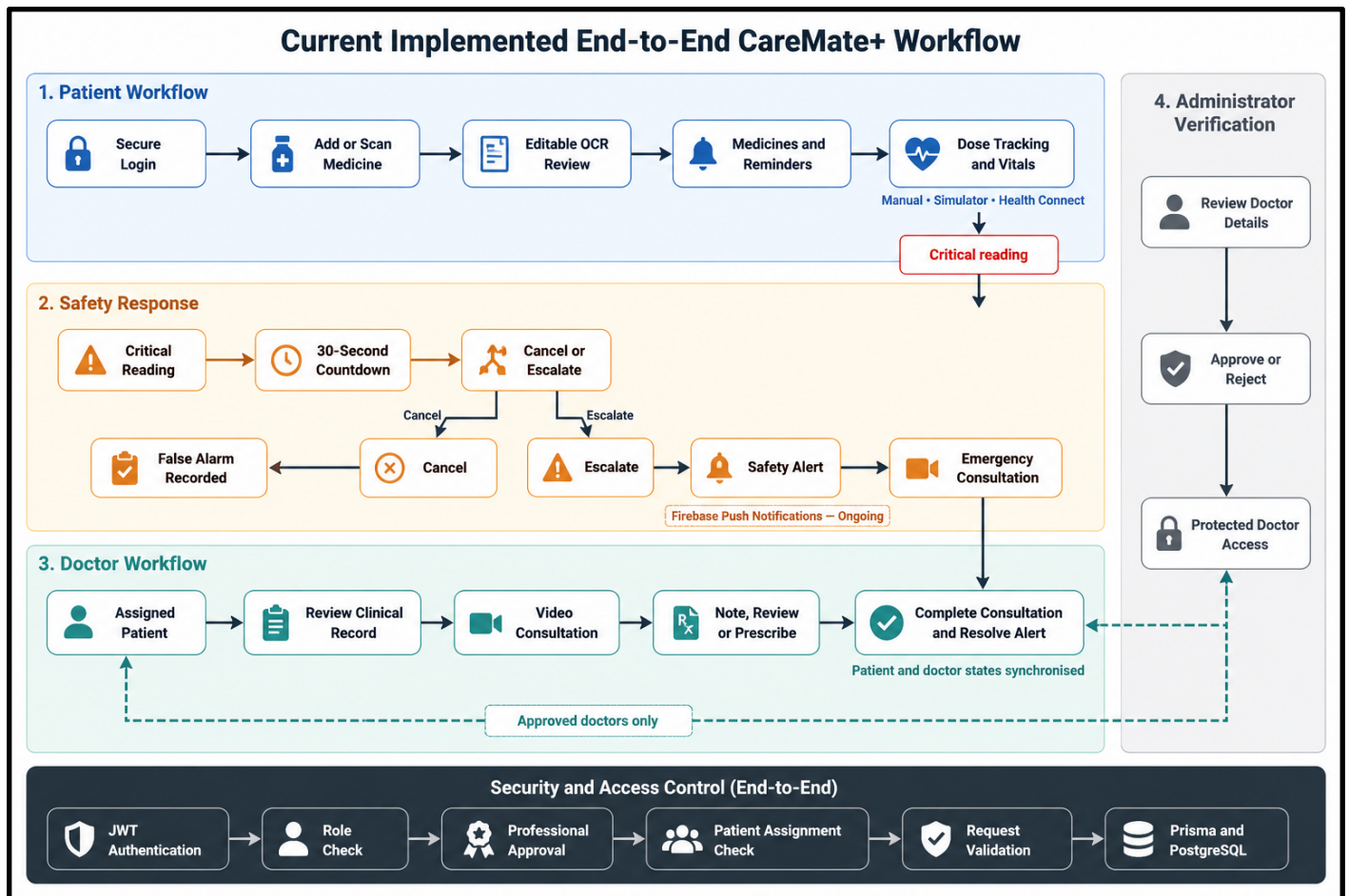


Figure 1: Current implemented end-to-end CareMate+ workflow.

2. CareMate+ Implementation Progress and Requirement Traceability Against the Preliminary Report

The SEC, PAT, DOC, ADM, CAR, PHA, REC and OPT requirement identifiers used in this section correspond to those defined in the preliminary report, enabling direct comparison between the original requirements and their implementation status at the interim stage.

2.1 Essential Security and Patient Requirements

Requirement ID	Requirement area	Implementation achieved	Status
SEC-01	Multi-role registration and login	Patient, doctor and administrator authentication flows are implemented. Caregiver and pharmacy account integration will be completed with their scheduled modules.	Ongoing
SEC-02– SEC-03	Authentication and role-based access	Passwords are securely hashed, APIs are protected using JWT authentication, and middleware prevents access outside each authorised role.	Complete
SEC-04– SEC-05	Relationship and professional-access controls	Patient ownership, active doctor-assignment checks and administrator approval of doctor accounts are implemented. Caregiver-link and pharmacy-approval checks will be added with the supporting modules.	Ongoing
SEC-06	Relational data persistence	PostgreSQL and Prisma relationships persist users, profiles, assignments, medicines, reminders, dose logs, vitals, alerts, consultations, notes, prescriptions and orders.	Complete
SEC-07	Audit of sensitive actions	Feature-specific timestamps and histories are stored. A central audit mechanism for verification, prescriptions, alerts and order changes remains under development.	Ongoing

Requirement ID	Requirement area	Implementation achieved	Status
PAT-01–PAT-02	Dashboard and medicine management	The dashboard presents medicines, reminders, vitals, consultations and doctor information. Patients can add, view, edit and delete medicines and use Today, Tomorrow and Week views.	Complete
PAT-03–PAT-04	OCR-assisted medicine capture	Camera or gallery images are processed using Google ML Kit. Recognised names, strengths, doses, frequencies and instructions remain editable and require confirmation before saving.	Complete
PAT-05	Reminder and adherence tracking	Reminder schedules and persistent dose logs support taken, missed and snoozed outcomes for scheduled medicines.	Complete
PAT-06	Vitals monitoring and Safety Response	Manual, simulated and Health Connect readings are classified as stable, warning or critical. Critical readings activate countdown, cancellation or escalation.	Complete

Table 2: Essential Security and Patient Requirement Progress

2.2 Essential Doctor and Administration Requirements

Requirement ID	Requirement area	Implementation achieved	Status
DOC-01	Approved and assigned-patient access	Only verified and administrator-approved doctors can enter protected clinical workflows. Active assignment checks restrict doctors to authorised patients and support multiple primary or specialist assignments.	Complete
DOC-02	Assigned-patient clinical review	Doctors can review patient profiles, medicines, reminder schedules, adherence records, latest and recent vital readings, scans, notes, Safety Alerts and consultation history.	Complete
DOC-03–DOC-04	Prescription creation and confirmation	Doctors can create prescriptions manually or upload images for OCR-assisted extraction. Multiple medicine items remain editable and require confirmation before linked prescription, medicine and reminder records are created.	Complete
DOC-05	Clinical coordination and consultation management	Doctors can manage medicine-review requests, add notes, accept or reject consultations, join Jitsi/JaaS video calls, handle Safety Alerts and complete emergency consultations. Completion synchronises both devices and resolves the linked alert.	Complete
ADM-01–ADM-02	Professional verification and access approval	Administrators can review doctor information and verification documents, approve or reject applications, and control protected access through persisted professional-account statuses.	Complete

Table 3: Essential Doctor and Administration Requirement Progress

2.3 Essential Caregiver and Pharmacy Requirement Progress

Requirement ID	Requirement area	Implementation achieved / planned delivery	Status
CAR-01–CAR-02	Linked-patient caregiver support	The caregiver requirements, access boundaries and usability workflow have been defined. The authenticated caregiver module will provide linked-patient medicines, reminders, missed doses, vital readings, alerts, consultation status and pharmacy-order visibility after the core patient, doctor and administrator modules are finalised.	Scheduled
PHA-01–PHA-02	Pharmacy order fulfilment	The pharmacy role, MedicineOrder data foundation and order lifecycle have been defined. The scheduled module will allow pharmacy staff to review prescription, refill and medicine orders, accept or reject requests, and update them through pending, accepted, preparing, ready and completed states.	Scheduled

Table 4: Essential Caregiver and Pharmacy Requirement Progress

2.4 Recommended and Optional Requirement Progress

Recommended Requirements

Requirement ID	Requirement area	Implementation achieved / remaining delivery	Status
REC-01–REC-03	Account recovery and confirmed OCR persistence	Email verification, time-limited password reset, and confirmation of OCR-derived medicine and reminder records are implemented.	Complete
REC-04	Sharing scans with assigned doctors	Prescription records and related patient information are available within protected doctor workflows. A dedicated shared-scan history view remains under development.	Ongoing
REC-05	Patient adherence history	Persisted taken, missed and snoozed dose logs are available through medicine and adherence views.	Complete
REC-06–REC-07	Refill requests and patient care diary	These extensions are scheduled alongside the pharmacy-order workflow and remaining patient-support features.	Scheduled
REC-08–REC-10	Doctor reports, scan advice and follow-up actions	Patient detail and doctor notes provide the underlying information. Dedicated reports, scan-linked advice and follow-up records remain under development.	Ongoing
REC-11–REC-13	Caregiver notes, consultation requests and pharmacy instructions	These requirements will be implemented with the scheduled caregiver and pharmacy modules.	Scheduled
REC-14	Administrator dashboard summaries	The administrator dashboard presents professional-verification information, application statuses and management actions.	Complete

Table 5: Recommended Requirement Progress

Optional Requirements

Requirement ID	Requirement area	Implementation achieved / remaining delivery	Status
OPT-01	Android Health Connect integration	Permission-controlled readings are synchronised to the backend and processed through the patient vitals and Safety Response workflows.	Complete
OPT-02	External medicine API	Repeated HTTP 503 responses led to the use of a local medicine-reference catalogue; live integration is deferred.	Deferred
OPT-03–OPT-04	Doctor analytics and trend views	Basic adherence and recent-vital information is available. Advanced filters, analytics and longer-term trends remain under development.	Ongoing
OPT-05–OPT-07	Heatmaps, text analysis and cloud deployment	These stretch features are deferred to prioritise supporting-role implementation, safety testing, usability evaluation and final evidence.	Deferred

Table 6: Optional Requirement Progress

2.5 Overall Requirement Status Summary

Status	Requirement areas	Current position
Complete	Core authentication; patient medicines, reminders, adherence, OCR, vitals and Safety Response; doctor assignments, clinical review, medicine reviews, prescriptions, consultations and alerts; administrator verification; Android Health Connect.	The principal patient-doctor-administrator pathway is implemented and integrated.
Ongoing	Central audit logging, Firebase push notifications, shared-scan history, advanced doctor reports and trends, ethics finalisation, formal testing and evaluation preparation.	Implementation or verification is active and included in the revised delivery plan.
Scheduled	Caregiver module, pharmacy fulfilment module, refill requests, patient care diary and supporting-role extensions.	Development will begin after the core modules are fully finalised and verified.
Deferred	NHS Medicines API integration, heatmaps, text analysis and cloud deployment.	The NHS endpoint repeatedly returned HTTP 503 errors, so a local medicine catalogue is used and other stretch features are deferred.

Table 7: Overall Requirement Status Summary

3. Technical Achievement Snapshot

CareMate+ combines several technically demanding components into one coordinated healthcare-support workflow. Its contribution is demonstrated through secure role relationships, OCR-assisted data transformation, health-data processing, safety-state management and synchronised patient-doctor actions.

3.1 Key MSc-Level Technical Contributions

Technical contribution	Implemented solution	MSc-level significance
Editable OCR medicine and prescription processing	Google ML Kit converts medicine or prescription images into structured drafts. Medicine name, strength, dosage, frequency, instructions and reminder times can be corrected before confirmation.	Combines text recognition, medicine-reference matching and human verification rather than storing unreliable raw OCR output.
Relationship-based clinical access control	JWT and role checks are combined with professional approval and active PatientDoctorAssignment validation before a doctor can access patient information.	Extends basic role-based security with database-enforced relationships and assigned-patient authorisation.
Health-data acquisition and classification	Manual, simulated and Android Health Connect readings are normalised, persisted and classified as stable, warning or critical.	Integrates multiple data sources into a consistent monitoring pipeline that drives later clinical workflows.
State-controlled Safety Response	Critical readings start a 30-second countdown with cancellation, immediate escalation or automatic escalation, creating a Safety Alert and emergency consultation.	Implements a time-sensitive, state-based workflow with persisted outcomes and safeguards against false escalation.
Prescription-to-reminder integration	A doctor can manually create or scan a multi-item prescription. Confirmed items create linked prescription, medicine and reminder records for the patient.	Coordinates OCR review, validation and related database writes across several healthcare entities.
Cross-device consultation lifecycle	Patient and doctor join the same Jitsi/JaaS consultation using role-specific access. Completion synchronises both devices and resolves the linked Safety Alert.	Demonstrates coordination between independent mobile clients, backend status management and linked workflow completion.

Table 8: Key MSc-Level Technical Contributions

These contributions show that the project’s technical depth lies in integrated workflow logic, security controls and coordinated data-state transitions rather than in screen quantity alone.

3.2 Architecture, Integration and Development Evidence

CareMate+ uses a layered architecture connecting role-specific React Native interfaces to protected Express and TypeScript REST endpoints. Requests pass through JWT authentication, role validation, professional-approval checks and active patient-assignment controls before domain services process medicines, vital readings, Safety Alerts, consultations, prescriptions and account verification. Prisma manages the relationships and state changes persisted in PostgreSQL.

Google ML Kit, Android Health Connect and Jitsi/JaaS are integrated through the relevant service workflows rather than being accessed directly from unrestricted screens. Implementation evidence includes real-device demonstrations, Postman API tests, Prisma Studio records, OCR samples, Health Connect synchronisation and two-device consultation testing.

Development work is committed regularly to dedicated GitLab feature branches rather than being merged immediately into the main branch. This approach keeps the main branch stable while incomplete or interdependent features are developed and verified. Completed and tested changes will be merged into the main branch after the relevant workflows have been finalised. The branch history therefore provides a traceable record of incremental development and testing, consistent with the module’s requirement to use GitLab regularly as evidence of progress. Backend API testing is documented through organised Postman screenshots covering authentication, medicines, reminders, vitals, Safety Response, consultations, OCR scanning and doctor-administrator workflows.

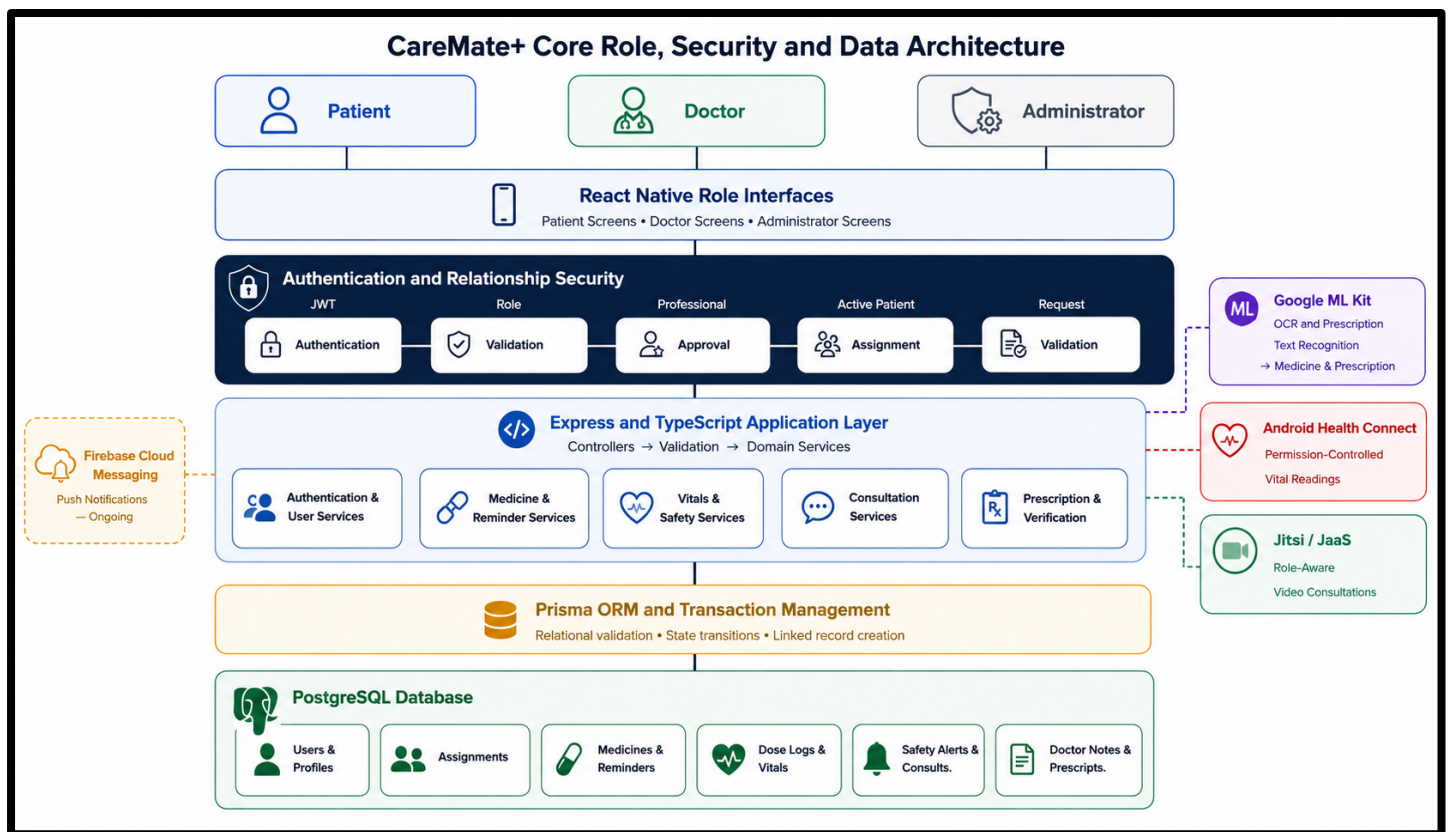


Figure 2: CareMate+ Core Role, Security and Data Architecture

4. Revisions to the Preliminary Plan and Evaluation Progress

4.1 Scope and Priority Changes

The preliminary plan proposed five user roles: Patient, Doctor, Administrator, Caregiver and Pharmacy. During implementation, the delivery order was revised so that the Patient–Doctor–Administrator pathway could be completed and verified first. This reduced integration risk because medicine management, vital monitoring, Safety Alerts, doctor assignments, consultations, prescriptions and professional approval depend on a stable shared backend and database model. Android Health Connect, OCR-assisted prescription processing and the complete Safety Response lifecycle were prioritised because they strengthen the project’s technical contribution and support the main care journey. Firebase push notifications remain under active development, while Caregiver access, Pharmacy fulfilment and refill workflows are scheduled for the remaining development period. The five-role objective has therefore been retained but phased to protect the stability of the core system.

NHS Medicines API integration was deferred after the tested endpoint repeatedly returned HTTP 503 Service Unavailable. CareMate+ therefore uses a locally seeded medicine-reference catalogue to keep medication and OCR workflows reliable. Heatmap visualisation, automated text analysis and cloud deployment were also deferred because they are non-essential stretch features. This revised scope protects the time required for integration testing, ethics approval, formal evaluation and completion of the essential workflows, while retaining the deferred features as possible future extensions.

4.2 Technical Challenges and Engineering Decisions

Several integration challenges affected the original implementation sequence. The Jitsi React Native SDK caused dependency and Android build conflicts, so Jitsi/JaaS consultations were temporarily opened through a browser or WebView to preserve a functional video-consultation pathway. Two-device testing required network configuration and adb reverse because university Wi-Fi and firewall restrictions prevented reliable access to the local backend. Prisma schema changes supporting multiple doctor assignments also caused migration drift, which was resolved by rebuilding the development database and rerunning the seed data. Duplicate Safety Response triggers from Health Connect and simulated critical readings were prevented through persisted alert-state checks and cancellation controls. In addition, repeated HTTP 503 responses from the NHS medicines endpoint led to the use of a locally seeded medicine-reference catalogue.

Engineering decisions prioritised a stable end-to-end prototype over forcing unreliable integrations. Patient data access is protected through role, account-approval and assignment checks rather than role validation alone. OCR results are shown as editable drafts before medicine or prescription records are created, reducing the risk of incorrect data being saved. Active development is completed in dedicated feature branches so incomplete changes do not destabilise the main branch. These decisions improved reliability while preserving Firebase notifications, native Jitsi integration and live NHS medicine integration as areas for later refinement.

4.3 Evaluation Completed So Far

Technical evaluation has been conducted across the mobile application, backend API and PostgreSQL database. The GitLab testing-evidence folder contains organised Postman screenshots covering health checks, registration, email verification, authentication, password recovery, medicine CRUD operations, reminder actions, stable, warning and critical vital readings, dashboard retrieval, Safety Alert creation, cancellation and escalation, manual and emergency consultations, OCR medicine scanning, multi-medicine prescription scanning, doctor registration and administrator approval. Positive workflow tests are supported by negative cases such as missing authentication tokens and failed login using an outdated password.

Further testing has been completed through Prisma Studio record inspection, real Android-device demonstrations and two-device patient–doctor consultation trials. These checks verified database state changes, Safety Response cancellation and escalation, consultation joining, repeated consultation use and synchronised completion. The evidence has been committed to dedicated GitLab development branches while the main branch is kept stable until the integrated workflows are finalised and verified. Representative screenshots will be included in the report, while the complete evidence set remains available in the GitLab docs/testing-evidence/postman-screenshots directory.

4.4 Ethics and Remaining Evaluation Work

Formal participant evaluation has not yet started because the ethics approval process is still being finalised. The participant information sheet, consent form, study protocol and questionnaire materials have been prepared, while supervisor-requested amendments are currently being incorporated. After these changes, the Infonetica application and supporting documents will be submitted for review. Human-participant usability testing will begin only after approval has been granted.

The remaining technical evaluation will include final regression testing, additional authorisation and boundary cases, Firebase notification verification, caregiver and pharmacy workflow testing, and final two-device consultation checks. Performance, usability and reliability findings will be analysed in the final report alongside Postman evidence, Prisma database records, real-device demonstrations and GitLab development history. Representative results will be included in the report, while the complete evidence set will remain in the repository.

5. Revised Project Plan

5.1 Comparison with the Original Schedule

Comparison with the preliminary schedule shows that the central Patient-Doctor-Administrator pathway is ahead of the original implementation plan. Safety Response escalation, prescription-to-reminder processing, Jitsi/JaaS consultation integration and initial API and device testing were achieved earlier than their planned completion windows. Caregiver and Pharmacy workflows are not currently delayed because their original implementation periods begin in Weeks 8 and 9.

Workstream	Original target	Position at interim stage	Schedule variance	Revised action
Patient medicines and reminders	Week 5	Core medicine, reminder and dose-state workflows completed and tested.	On schedule / slightly early	Final regression and edge-case testing.
Safety Response escalation	Week 7	Countdown, cancellation and escalation implemented by Week 6.	1 week early	Complete Firebase notification delivery and final verification.
Doctor prescriptions and reminder creation	Weeks 7-8	Manual and scanned prescription workflows substantially implemented by Week 6.	1-2 weeks early	Complete integration and boundary-case testing.
Jitsi/JaaS consultation integration	Week 10	Two-device patient-doctor consultation workflow available by Week 6.	Approx. 4 weeks early	Refine lifecycle reliability and consolidate evidence.
API, database and device evidence	Week 11	Postman, Prisma Studio and real-device evidence collection already underway.	Started early	Continue regression testing and organise representative evidence.
Caregiver module	Week 8	Not yet started; original implementation window has not begun.	On schedule	Begin during the revised Week 8 window.
Pharmacy module	Week 9	Not yet started; original implementation window has not begun.	On schedule	Begin after Caregiver integration.
Ethics and participant evaluation	Weeks 8-11	Supporting documents prepared; supervisor amendments are ongoing.	On schedule	Submit through Infonetica and evaluate after approval.

Table 9: Original and Actual Schedule Comparison

The revised plan reallocates the available schedule margin to Caregiver and Pharmacy development, Firebase notifications, audit logging, final regression testing and formal evaluation rather than expanding the scope with additional optional features.

5.2 Updated Twelve-Week Schedule

The revised schedule retains the original final submission deadline while reflecting progress achieved by the interim stage. The central Patient–Doctor–Administrator pathway is approximately one to two weeks ahead of the original plan, as Safety Response escalation, prescription-to-reminder processing and Jitsi/JaaS consultation integration were achieved earlier than scheduled, while testing-evidence collection also began earlier than planned. The resulting schedule margin is therefore reallocated to Caregiver and Pharmacy development, Firebase notifications, central audit logging, technical hardening, final regression testing and formal participant evaluation subject to ethics approval.

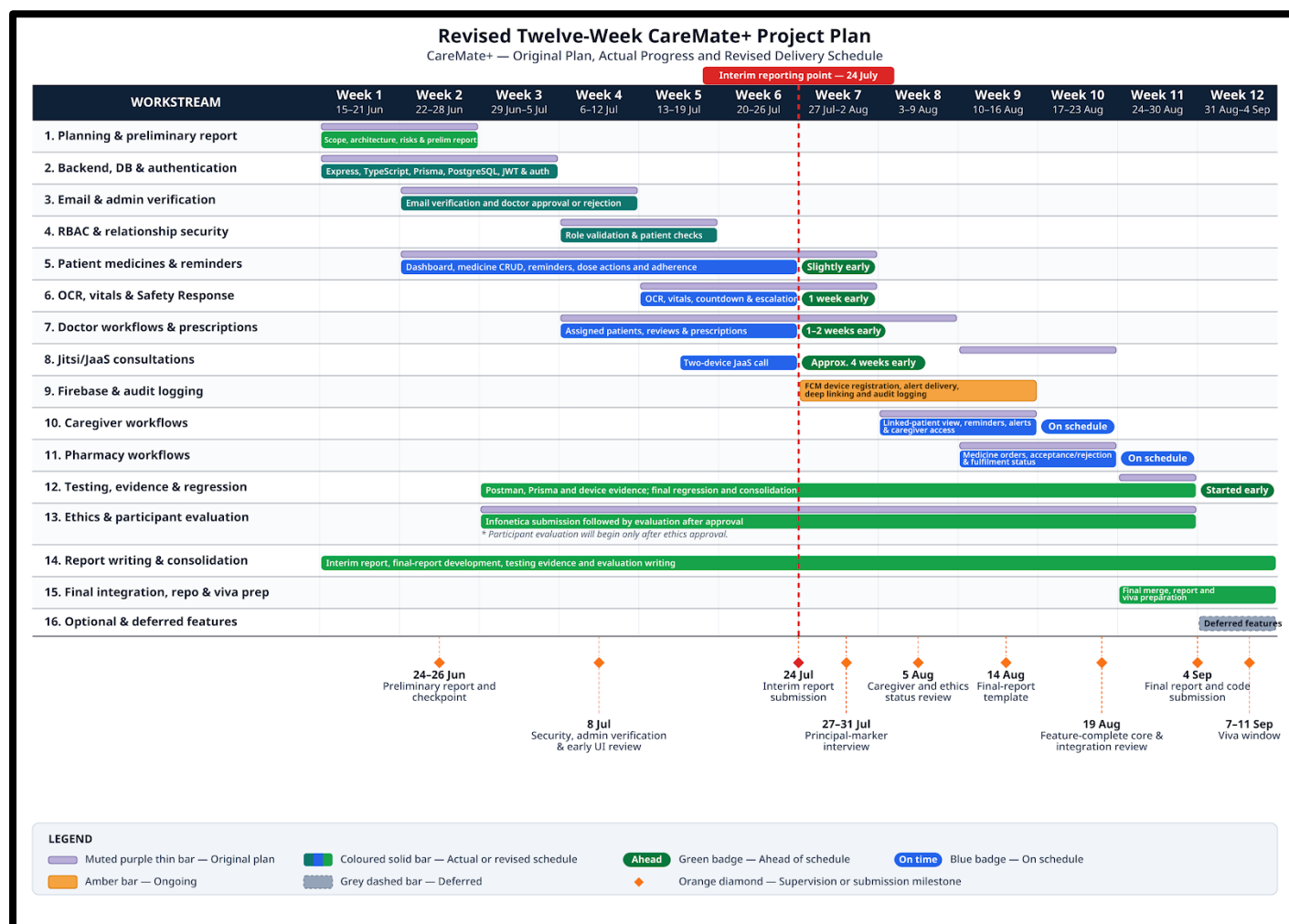


Figure 3: Revised twelve-week CareMate+ project plan (Gantt chart) showing the original schedule, actual progress and revised delivery plan.

Week	Date range	Revised implementation focus	Delivery position
1	15–21 Jun	Project scope, repository structure, backend foundation, database planning and initial architecture.	Complete
2	22–28 Jun	Authentication, JWT, email verification, role profiles, preliminary report and early patient interfaces.	Complete
3	29 Jun–5 Jul	Prisma/PostgreSQL integration, administrator approval workflow, patient dashboard, medicines and reminders.	Complete
4	6–12 Jul	RBAC, professional verification, patient-assignment checks, OCR medicine scanning, vitals and Health Connect.	Complete
5	13–19 Jul	Safety Response countdown and cancellation, doctor assigned-patient workflows, medicine review and consultations.	Complete
6	20–26 Jul	Escalation lifecycle, prescriptions, prescription-to-reminder integration, administrator integration, two-device testing and interim evidence.	Current – ahead

Week	Date range	Revised implementation focus	Delivery position
7	27 Jul–2 Aug	Firebase notification delivery, central audit logging, remaining core refinements and principal-marker preparation.	Ongoing
8	3–9 Aug	Caregiver module, linked-patient access, reminders, alerts and relationship-based access controls.	Scheduled
9	10–16 Aug	Pharmacy fulfilment, refill and order-status workflows, plus final-report template preparation.	Scheduled
10	17–23 Aug	Caregiver and Pharmacy workflow completion, Firebase and audit-log finalisation, security hardening, boundary-case checks and performance refinement.	Scheduled
11	24–30 Aug	Ethics completion, participant evaluation where approved, systematic testing evidence, result consolidation and evaluation writing.	Scheduled
12	31 Aug–4 Sep	Final bug fixes, report completion, repository clean-up, final branch integration, demonstration preparation and viva preparation.	Scheduled

Table 10: Revised Weekly Implementation and Delivery Plan

The schedule does not treat Caregiver and Pharmacy work as delayed, because their original implementation windows begin after the interim point. Early completion of core clinical workflows is used as contingency for integration, evaluation and final-quality assurance rather than for uncontrolled scope expansion.

5.3 Remaining Milestones and Risk Mitigation

The remaining delivery period will focus on completing Firebase notifications and central audit logging, followed by the Caregiver and Pharmacy workflows, cross-role integration, regression testing and final evaluation. Incomplete features will continue to be developed in dedicated GitLab branches and will be merged into the main branch only after verification. Essential requirements will remain the priority, while deferred optional features will not be reintroduced unless sufficient schedule margin remains.

The final milestones are completion of the supporting-role workflows, full regression and security testing, ethics submission through Infonetica, participant evaluation after approval, evidence consolidation, final report completion, repository clean-up and viva preparation. The main risks and controls are summarised below.

Risk	Potential impact	Control and fallback
Caregiver or Pharmacy development takes longer than planned	Less time remains for cross-role integration, testing and report completion.	Implement essential workflows first, reuse existing authentication and relationship services, and keep non-essential supporting features outside the delivery scope.
New modules destabilise the completed Patient-Doctor-Administrator pathway	Previously working medicine, alert, consultation or prescription workflows may fail.	Develop in separate feature branches, run API and real-device regression tests, and merge into the main branch only after verification.
Firebase or third-party integration problems	Push notification delivery or consultation behaviour may remain incomplete.	Isolate integrations behind service layers, retain working in-app alerts and Jitsi/JaaS fallback workflows, and document any remaining platform limitation.
External service unavailability	Medicine, notification or consultation features may be disrupted.	Use isolated services and fallbacks: a local medicine catalogue replaces the unavailable NHS endpoint, while in-app alerts and Jitsi/JaaS preserve core workflows.
Ethics approval is delayed	Participant usability evaluation cannot begin within the planned period.	Complete amendments and submit through Infonetica promptly. Do not involve participants before approval; use a supervisor-approved non-participant technical evaluation fallback if required.
Insufficient time for final testing and report completion	Defects, weak evaluation evidence or incomplete documentation may remain.	Freeze optional development, reserve the final weeks for regression testing and writing, and continue collecting evidence throughout implementation.
Loss or instability of development work	Progress may be delayed or difficult to demonstrate.	Maintain regular GitLab commits, preserve working versions in feature branches, and retain organised Postman, database and device-testing evidence.

Table 11: Remaining Project Risks and Controls

These controls protect the completed core workflow while ensuring that the remaining functionality, evaluation and documentation can be delivered within the final project period.